Worksheet 4-7 A L'Hospital's Rule #1
AP Calculus Name_____

DO ALL WORK ON NOTEBOOK PAPER.
Directions: Find the limit. Use L'Hospital's Rule where appropriate. If there is a more elementary method, consider using it.

1. $\lim\limits_{x\to 3}\dfrac{2x-6}{x^2-9}$

2. $\lim\limits_{x\to 3}\dfrac{\sqrt{x+1}-2}{x-3}$

3. $\lim\limits_{x\to\infty}\dfrac{5x^2-3x+1}{3x^2-5}$

4. $\lim\limits_{x\to 2}\dfrac{x^3-x-2}{x-2}$

5. $\lim\limits_{x\to 0}\dfrac{e^x-(1-x)}{x}$

6. $\lim\limits_{x\to 0}\dfrac{\sin(2x)}{\sin(3x)}$

7. $\lim\limits_{x\to\infty}\dfrac{3x^2-2x+1}{2x^2+3}$

8. $\lim\limits_{x\to\infty}\dfrac{x^2+2x+1}{x-1}$

9. $\lim\limits_{x\to\infty}\dfrac{x}{\sqrt{x^2+1}}$

10. $\lim\limits_{x\to\infty}\dfrac{\ln x}{x}$

11. $\lim\limits_{x\to\infty}\dfrac{x^2}{e^{5x}}$

12. $\lim\limits_{x\to 0}\dfrac{\cos x-1}{e^x-x-1}$

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13. Functions f, g, and h are twice-differentiable functions with $g(3)=h(3)=5$. The line $y=5+\frac{1}{2}(x-3)$ is tangent to both the graph of g at $x=3$ and the graph of h at $x=3$.

a.) Find $h'(3)$

b.) Let a be the function given by $a(x)=2x^3h(x)$. Write an expression for $a'(x)$. Find $a'(3)$.

c.) The function h satisfies $h(x)=\frac{x^2-9}{1-(f(x))^3}$ for $x\neq 3$. It is known that $\lim\limits_{x\to 3}h(x)$ can be evaluated using L'Hospital's Rule. Use $\lim\limits_{x\to 3}h(x)=5$ to find $f(3)$ and $f'(3)$. Show the work that leads to your answers.

1.) $\frac{1}{3}$	2.) $\frac{1}{4}$	3.) $\frac{5}{3}$	4.) DNE	5.) 2	6.) $\frac{2}{3}$	7.) $\frac{3}{2}$
8.) $+\infty$	9.) 1	10.) 0	11.) 0	12.) -1		

Directions: Evaluate the following limits, if possible. Then circle the correct answer.

14. $\lim_{x \to 0} \dfrac{\cos(2x + \pi) + x + 1}{e^{x^2} + 2x - 1}$

(A) $\dfrac{1}{3}$ (B) $\dfrac{1}{2}$ (C) $\dfrac{2}{3}$ (D) does not exist

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$\boldsymbol{x}$	-7	-5	-3
$\boldsymbol{f(x)}$	0	-3	0
$\boldsymbol{f'(x)}$	6	-1	9

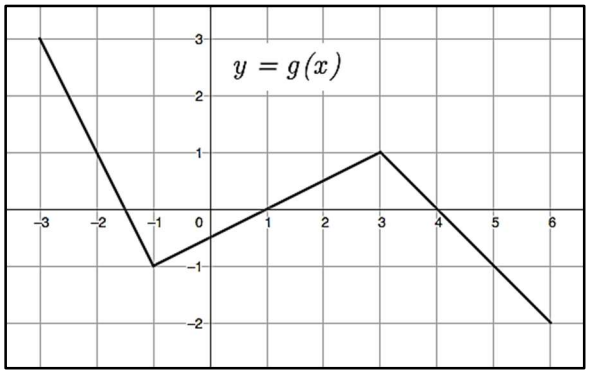
The table above gives selected values of a twice-differentiable function $f(x)$. Use the table to answer questions $15 - 16$.

15. $\lim_{x \to -3} \dfrac{f(2x - 1)}{x^2 - 9}$

(A) -3 (B) -2 (C) -1 (D) does not exist

16. $\lim_{x \to -5} \dfrac{f(f(x))}{3x + 15}$

(A) $\dfrac{1}{3}$ (B) -3 (C) 3 (D) does not exist



17. The graph of $g(x)$is above. Use the graph to find $\lim_{x \to 4} \dfrac{g(x - 3)}{x^2 - 2x - 8}$ if it exists.

(A) $\dfrac{1}{12}$ (B) $-\dfrac{1}{6}$ (C) 3 (D) does not exist